

OpenCPU SDK

WiFi API Notes

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Notice

This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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Contents

1 Overview	5
2 API Introduction	6
2.1 Data Structure	6
2.2 API	6
2.2.1 nwy_wifi_scan	6

About This Document

Scope

This document is applicable to N706B and N520 modules.

Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Li Xuanting

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

WiFi scanning is primarily used to detect available wireless networks nearby. Specifically, it helps users identify networks within the area to select the best one for connection. During a WiFi scan, the device sends or listens for specific signal frames to gather information about nearby wireless networks, such as: Network Name (SSID), Signal Strength, and Encryption Type.

This document introduces WiFi scanning interfaces, aiding developers in quickly implementing related applications.

2 API Introduction

2.1 Data Structure

Maximum number of WiFi scan results:

```
#define NWY_WIFI_SCAN_AP_MAX 16
```

Structure for a single WiFi access point's information:

```
typedef struct
```

```
{
```

```
    uint8 mac [6]; /* Array to store the MAC address of a WiFi access point. A MAC address is a 6-  
                    byte (48-bit) identifier uniquely identifying a device on a network. Typically  
                    represented in hexadecimal and divided into 6 groups, each with two digits,  
                    e.g., XX:XX:XX:XX:XX:XX. */
```

```
    uint8 channel; /* The channel number used by the WiFi access point. */
```

```
    int rssi; /* Received signal strength indicator (RSSI). */
```

```
    uint8_t ssid[32];
```

```
} nwy_wifi_ap_info_t;
```

Wifi scan results:

```
typedef struct
```

```
{
```

```
    int num; /*The number of scanned WiFi's*/
```

```
    nwy_wifi_ap_info_t ap_list[NWY_WIFI_SCAN_AP_MAX]; /*Stores the scanned WiFi  
information*/
```

```
} nwy_wifi_scan_list_t;
```

2.2 API

The API definitions can be found in **nwy_wifi_api.h**.

2.2.1 nwy_wifi_scan

Function	nwy_error_e nwy_wifi_scan(nwy_wifi_scan_list_t *scan_list);
Description	WiFi scanning interface
Parameter	Input parameter: scan_list: Used to store WiFi scanning information.

Return Value	Output parameter: scan_list: Contains the details of the scanned WiFi results.
	Successful: NWY_SUCCESS Failure: Refer to nwy_error_e for error codes.
Notes	The WiFi scanning function works properly when both LTE and GNSS (Global Navigation Satellite System) are inactive. Under normal conditions, scanning is completed within 1–5 seconds, and the scanning process automatically stops afterward. The maximum timeout for a normal scan is 30 seconds.